Uniqueness of nonnegative heat controls at the minimum time

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Abstract

We consider the heat equation on an interval with two nonnegative Dirichlet boundary controls. For every pair of distinct positive constant states, we prove that the nonnegative finite Radon measure pair attaining the minimum time is unique, under the terminal sine-moment formulation. The two boundary measures are equal. This answers the continuum uniqueness question posed by Lohéac, Trélat and Zuazua. The proof uses finite positive moment cones at the critical time. Their extremal supporting functions have vanishing cost on every admissible measure. Exponential interpolation changes their exponents without increasing their magnitude, while a determinant estimate keeps an inserted coefficient away from zero. These tests determine both the average control and the difference between the boundary controls, including endpoint atoms.

1 Introduction

Nonnegative boundary temperatures constrain how quickly a rod can be heated or cooled. We study the heat equation

$$\begin{cases} y_t = y_{xx}, & t > 0, \quad 0 < x < 1, \\ y(0,x) = a, & \\ y(t,0) = u_0(t), \qquad y(t,1) = u_1(t), & \end{cases} \tag{1}$$

where $a, b > 0$, $a \neq b$, and the target is $y(H,x) = b$. There is no upper bound on the permitted nonnegative boundary temperatures, but each classical control is bounded on its time interval. Write

$$T_*(a,b) = \inf\{H > 0 : \text{bounded } u_0, u_1 \geq 0 \text{ take } a \text{ to } b \text{ at time } H\}. \tag{2}$$

Without the nonnegativity constraint, bounded null controls exist on every positive duration. Positivity changes the minimum-time question: Lohéac, Trélat and Zuazua proved a finite, strictly positive waiting time and attainment by nonnegative measure controls [1, Theorem 2.1].

The remaining question is whether the fastest measure control is uniquely determined. It was posed in [1, Section 6], with the terminal state expressed through sine moments. Further continuum structure was obtained by Lohéac for a model with one controlled endpoint and a homogeneous condition at the other [2, Theorem 1]. For reachable steady-state targets generated by a positive boundary control, that work constructs a countably impulsive minimum-time control and proves uniqueness within the purely impulsive class. Uniqueness among all Radon controls is left open there [2, Section 5]. For finite-dimensional systems with real spectral hypotheses, uniqueness was proved

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in [3]. They include discretized heat systems, but finite-dimensional uniqueness does not identify the limiting pair for the continuum equation. Here we prove uniqueness for the full two-boundary measure pair directly from all its moments.

We use the closed-interval measure formulation to state precisely what attainment means. A pair (μ_0, μ_1) of finite nonnegative Radon measures on $[0, H]$ is admissible if, for every integer $k \geq 1$,

$$\int_{[0,H]} e^{-k^2\pi^2(H-t)}\bigl(d\mu_0(t) - (-1)^k d\mu_1(t)\bigr) = \frac{(1-(-1)^k)(b - ae^{-k^2\pi^2 H})}{k^2\pi^2}. \tag{3}$$

These are the terminal-state sine equations of (1). For bounded controls the measures are $u_i(t)\,dt$; the definition also retains possible atoms at zero and at H. It is the explicit moment convention in AIM Problem 138[1], so the theorem addresses that measure question without adding a terminal weak-solution convention.

Theorem 1.1. *For every $a, b > 0$ with $a \neq b$, $0 < T_*(a,b) < \infty$. At $H = T_*(a,b)$ there exists exactly one nonnegative finite Radon pair on $[0,H]$ satisfying (3) for all $k \geq 1$. Its components coincide: $\mu_0 = \mu_1$.*

The existence and positive waiting time are inherited from the established control theory after the convention bridge proved below. The new conclusion is uniqueness of the complete pair. In particular, symmetry of the equation is forced on every admissible minimum-time pair, rather than imposed on the controls at the outset.

Two issues obstruct a direct moment argument. The average boundary measure is prescribed only at the odd heat eigenvalues, whose reciprocals have a finite sum; ordinary density of all powers cannot be used to determine it. Even uniqueness of this average would leave open whether an asymmetric pair has a signed difference with all even moments zero. Criticality supplies the additional information needed for both questions. A positive interval of density would allow the controls to be advanced slightly in time and corrected by a small bounded null control, contradicting minimality.

At the fixed critical time, the finite moment cones therefore approach an extremal initial-to-target ratio. Their nonnegative supporting exponential polynomials have integrals tending to zero against every critical measure. Their zeros exhaust the finite exponential zero bound. This permits us to insert a whole interval of new exponents while retaining pointwise domination by the original tests. Compactness results for sparse power polynomials [4], together with a normalized determinant estimate, ensure that the new coefficient does not vanish. The resulting transforms determine the average measure. Interlacing of odd and even eigenvalues lets the same interpolation determine the boundary difference.

The argument concerns distinct positive constant states on the interval. It gives neither a formula for the minimum time nor a uniqueness theorem for nonconstant profiles. Section 2 establishes the critical moment formulation and the no-density property. Section 3 proves the interpolation estimates. Section 4 constructs the extremal tests and proves the two rigidity statements. All measure integrals over a displayed time interval include its endpoints.

2 Critical heat moments

We write $\mathcal{M}_+([0,H])$ for the finite nonnegative Radon measures on the closed interval $[0,H]$. All signed measures below are real. After dividing the state and controls by b, put

$$r = \frac{a}{b}, \qquad p = \pi^2, \qquad \lambda_j = (2j-1)^2 p, \qquad \nu_j = (2j)^2 p \quad (j \geq 1). \tag{4}$$

[1] AIM Problem 138, pinned statement.

This scaling is invertible and preserves every feasible time. For a boundary pair define its average and difference by

$$w = \frac{\mu_0 + \mu_1}{2b}, \qquad h = \frac{\mu_0 - \mu_1}{2b}. \tag{5}$$

The pair is nonnegative precisely when $w \geq 0$ and $|h| \leq w$, where $|h|$ denotes total variation. Indeed, every Borel partition of a set A gives $\sum_\ell |h(A_\ell)| \leq w(A)$; taking the supremum proves the domination. Conversely $w \pm h \geq 0$ under that domination. The inverse pair is $\mu_0 = b(w + h)$, $\mu_1 = b(w - h)$.

Multiplying the terminal sine equations by the terminal exponential yields

$$\int_{[0,H]} e^{\lambda_j t}\, dw(t) = \frac{e^{\lambda_j H} - r}{\lambda_j}, \tag{6}$$

$$\int_{[0,H]} e^{\nu_j t}\, dh(t) = 0. \tag{7}$$

Thus uniqueness has two parts: uniqueness of the positive odd-moment representation, and vanishing of the dominated signed even-moment difference.

For reference, the unnormalized sine coefficient $y_k(t) = \int_0^1 y(t,x)\sin(k\pi x)\,dx$ satisfies

$$y_k(H) = e^{-k^2 pH} y_k(0) + k\pi \int_0^H e^{-k^2 p(H-t)} \big(u_0(t) - (-1)^k u_1(t)\big)\, dt. \tag{8}$$

We use the sine convention and the bounded null-controllability result of Lohéac–Trélat–Zuazua [1, Section 2.2 and Lemma 2.1]: for every $L > 0$ and $z \in L^2(0,1)$, real C^1 boundary controls take z to zero in time L, with

$$\max_{i=0,1} \|v_i\|_\infty \leq \kappa(L)\, \|z\|_2 . \tag{9}$$

Their Theorem 2.1 also gives a finite positive infimum for nonnegative L^1 controls between distinct positive constants. The next lemma relates that convention to bounded controls and to the closed-interval moments.

Lemma 2.1. *The infima over bounded nonnegative controls, nonnegative L^1 controls, and nonnegative measure pairs satisfying the terminal moments agree. Their common value $T = T_*(a,b)$ lies in $(0,\infty)$, and the measure moments are attained on $[0,T]$.*

Proof. Suppose a nonnegative pair satisfies the moments at $H > 0$. Extend each measure to $\mathbb{R}$ by density a on $(-\infty, 0)$ and density b on (H, ∞), retaining its entire measure on $[0,H]$. Choose a nonnegative smooth kernel ρ with integral one and compact support in $(0,\epsilon)$. Convolution gives smooth bounded nonnegative controls on $[0, H+\epsilon]$.

For an odd rate λ let $\widetilde{u}$ be the extended average. For every s in the kernel support, its three contributions give

$$\begin{aligned} e^{\lambda s} \int_{[-s,H+\epsilon-s]} e^{\lambda t}\, d\widetilde{u}(t) &= \frac{e^{\lambda s}}{\lambda}\big[a(1 - e^{-\lambda s}) + (be^{\lambda H} - a) + b(e^{\lambda(H+\epsilon-s)} - e^{\lambda H})\big] \\ &= \frac{be^{\lambda(H+\epsilon)} - a}{\lambda}. \end{aligned}$$

Integrating against $\rho(s)\,ds$ proves the smoothed odd moments by Tonelli's theorem. The extended difference is zero off $[0,H]$, so each smoothed even moment is the original zero moment multiplied by $\int e^{\nu s}\rho(s)\,ds$. Fubini applies against finite total variation on the relevant compact interval. Both original endpoint atoms are included.

These smooth controls equal a near time zero and b near time $H+\epsilon$. To see that their moments identify an actual terminal state, subtract the affine lifting $F(t,x)=(1-x)u_0(t)+xu_1(t)$. The remainder has homogeneous boundary values and forcing $-F_t$. Its sine series defines an L^2 solution: partial sums obey

$$\|z(t)\|_2 \le \|z(0)\|_2 + \int_0^t \|F_s(s)\|_2 \, ds,$$

and the same bound for tails, followed by L^2 completeness, proves convergence. Formula (8) and completeness of the sine basis then give the constant terminal state b. Thus every measure-feasible time can be approximated from above by bounded-control times. Since bounded controls are L^1 controls and L^1 controls define measures with the same moments, all three infima agree. The cited positive finite waiting-time theorem applies to $a \neq b$.

Choose bounded admissible pairs at times $T_n \downarrow T$, with $T_n \le T+1$, and extend them by zero to $[0, T+1]$. Their averages u_n satisfy

$$u_n([0,T+1]) \le \int e^{pt}\, du_n(t) = \frac{be^{pT_n}-a}{p}.$$

Each component is bounded by $2u_n$, so the two positive measures have uniformly bounded mass. Weak-star sequential compactness on the common compact interval gives a limiting pair. Positivity passes through nonnegative continuous tests. Any compact subset of $(T, T+1]$ eventually lies beyond the supports, and hence has zero limiting mass; the limit is supported on $[0,T]$. Each fixed exponential is continuous, so all terminal moments pass to the limit. This proves attainment. □

From now on $T = T_*(a,b) > 0$. A *critical scalar measure* means a $w \in \mathcal{M}_+([0,T])$ satisfying every equation (6) at $H = T$. Such measures exist by Lemma 2.1.

Lemma 2.2. *A critical scalar measure cannot dominate $\delta\, dt$, $\delta > 0$, on an interval (α, β) with $0 < \alpha < \beta < T$.*

Proof. Suppose it does, and retain $m_0 = w(\{0\})$. For small $\epsilon > 0$ put $H = T - \epsilon$ and translate the part on $[\epsilon, T]$ to the left:

$$w^\epsilon = m_0\delta_0 + (t \mapsto t-\epsilon)_\#(w|_{[\epsilon,T]}).$$

The required odd moment at H minus its present value is

$$C_j^\epsilon = (e^{-\lambda_j\epsilon}-1)(r/\lambda_j + m_0) + e^{-\lambda_j\epsilon}\int_{(0,\epsilon)} e^{\lambda_j t}\, dw(t). \tag{10}$$

The discarded interval is open; the atom at ϵ is translated and the original atom at zero is kept. For each fixed j, $C_j^\epsilon \to 0$, and for all j, ϵ its magnitude is at most $r/p + m_0 + w([0,T])$.

Choose $0 < A < B < T$ strictly inside (α, β). For sufficiently small ϵ, $B < H$ and $w^\epsilon \ge \delta\, dt$ on $[A,B]$. Let $z_\epsilon \in L^2(0,1)$ have unnormalized sine coefficients

$$\int_0^1 z_\epsilon(x)\sin(k\pi x)\, dx = \begin{cases} -2k\pi C_j^\epsilon e^{-k^2pA}, & k = 2j-1, \\ 0, & k \text{ even}. \end{cases}$$

This is an actual real L^2 state, defined by the convergent orthogonal series. Parseval's identity and dominated convergence give

$$\|z_\epsilon\|_2^2 = 8\pi^2 \sum_{j\ge1} (2j-1)^2 |C_j^\epsilon|^2 e^{-2\lambda_j A} \longrightarrow 0. \tag{11}$$

Only odd sine modes occur, so this state is invariant under $x \mapsto 1-x$. Apply (9) on the fixed duration $B-A$. Averaging the two null controls leaves the odd forcing unchanged and makes the even forcing zero. The resulting common real control v_ϵ on $[A,B]$ has supremum norm tending to zero. Multiplying its zero terminal odd coefficient in (8) by $e^{\lambda_j B}/(2k\pi)$ gives

$$0 = -C_j^\epsilon + \int_A^B e^{\lambda_j t} v_\epsilon(t)\, dt.$$

Extend v_ϵ by zero outside $[A,B]$. Once $\|v_\epsilon\|_\infty \le \delta$, the measure $w^\epsilon + v_\epsilon\, dt$ is nonnegative and satisfies all the odd moments at H. Use it for both normalized boundary measures; the even moments then vanish. The smoothing in Lemma 2.1, with extra duration $\epsilon/2$, gives bounded controls at $T-\epsilon/2$. This contradicts the definition of T. □

3 Exponential interpolation

We need to change a finite test function's exponents while keeping its zeros and controlling its magnitude. Zeros are counted with multiplicity throughout this section.

Lemma 3.1. *A nonzero combination of n distinct real exponentials has at most $n-1$ real zeros. For increasing exponents $\alpha_1 < \cdots < \alpha_n$, every square confluent evaluation determinant is positive when rows are ordered by increasing real node and then by increasing derivative order, and columns by increasing exponent. For any root multiset of total multiplicity $n-1$, the interpolants in these n exponentials form a one-dimensional space; their coefficients are nonzero and alternate in sign. They have precisely the prescribed real roots and multiplicities.*

Proof. Factor out the smallest exponential and differentiate. Rolle's theorem, including the $m-1$ retained zeros at a root of multiplicity m, reduces the bound to $n-1$ exponentials. Induction bounds any finite set of roots, hence all real roots. Linear independence follows also from the Wronskian

$$e^{(\alpha_1+\cdots+\alpha_n)t} \prod_{i<j} (\alpha_j - \alpha_i) > 0.$$

The zero bound makes every square confluent evaluation matrix nonsingular. For distinct ordered nodes the determinant has constant sign on their connected ordering region. As all nodes tend to t, Taylor expansion gives

$$\frac{\det[e^{\alpha_i \tau_j}]_{i,j=1}^n}{\prod_{i<j}(\tau_j - \tau_i)} \longrightarrow \frac{e^{t\sum_i \alpha_i} \prod_{i<j}(\alpha_j - \alpha_i)}{\prod_{k=0}^{n-1} k!} > 0.$$

Partial coalescence replaces each block by successive derivative rows after division by its positive local Vandermonde and factorials. Its limit is nonzero by the same zero bound, and is therefore positive. For $n-1$ root conditions, all square cofactors are positive. The null vector thus has nonzero alternating coefficients. The global zero bound excludes further roots and higher multiplicities. □

Lemma 3.2. *Let q be an interpolant with n distinct real exponents and $n-1$ prescribed real roots. Fix one exponent α_k and its coefficient. Move another exponent α_j strictly toward α_k without crossing an exponent, and let P have the same roots and the same coefficient at α_k. Then*

$$|P(t)| \le |q(t)| \qquad (t \in \mathbb{R}). \tag{12}$$

Proof. The alternating coefficients in Lemma 3.1, together with the shared coefficient, give P and q the same sign near $+\infty$. Their exact root multiplicities agree, so their signs agree on every complementary interval.

The difference $d = P - q$ cancels the shared term. It uses n distinct exponents: the $n-2$ unchanged ones and the old and new moved exponents. The old exponent still has index j in this ordered list. If $j < k$, insertion and deletion occur above it; if $j > k$, deletion below and insertion just below it cancel their changes of index. Its coefficient in d is the negative of its coefficient in q. All $n-1$ prescribed zeros remain zeros of d. Coefficient alternation therefore gives d the opposite sign to q near $+\infty$, and then on each complementary interval. Thus P lies between zero and q there. All three vanish at the roots, proving (12). □

The magnitude bound alone is insufficient if the coefficient of the new exponential tends to zero. A normalized determinant estimate will prevent that loss. Write $V(z) = \prod_{i<j}(z_j - z_i)$ and set

$$\mathcal{H}_n(\alpha, \tau) = \frac{\det[e^{\alpha_i \tau_j}]_{i,j=1}^n}{V(\alpha)V(\tau)}.$$

It is positive for distinct real lists and symmetric in each list. At repeated nodes it has a continuous positive extension given by the confluent determinants above. Empty Vandermondes have value one.

Lemma 3.3. *For nodes in $[0,T]$, increasing one exponent by $d > 0$, while preserving distinctness, gives*

$$1 \le \frac{\mathcal{H}_n(\alpha + de_i, \tau)}{\mathcal{H}_n(\alpha, \tau)} \le e^{dT}. \tag{13}$$

The estimate also holds for repeated nodes.

Proof. For distinct ordered nodes a determinant integration gives the recursion

$$\mathcal{H}_n(\alpha, \tau) = \frac{1}{V(\tau)} \int_{\tau_1}^{\tau_2} \cdots \int_{\tau_{n-1}}^{\tau_n} V(u)\mathcal{H}_{n-1}(\alpha_1, \dots, \alpha_{n-1}; u) e^{\alpha_n(\sum_j \tau_j - \sum_i u_i)}\, du. \tag{14}$$

Here $\mathcal{H}_1(\alpha_1, \tau_1) = e^{\alpha_1 \tau_1}$, and any chosen exponent can occupy the last position. To verify the recursion, put $\beta_i = \alpha_i - \alpha_n$. Integrating the determinant $\det[e^{\beta_i u_j}]_{i,j<n}$ column by column gives the entries $(e^{\beta_i \tau_{j+1}} - e^{\beta_i \tau_j})/\beta_i$. In the original determinant, factor $e^{\alpha_n \tau_j}$ from column j, subtract consecutive columns and expand the last row $(1, 0, \dots, 0)$. The sign $(-1)^{n+1}$ cancels the sign in

$$\prod_{i<n} \beta_i = (-1)^{n-1} \prod_{i<n} (\alpha_n - \alpha_i).$$

Division by the two Vandermondes yields (14), with no additional factorial. Signed Vandermondes make this calculation valid for any chosen last exponent.

On the integration region,

$$0 \le \tau_1 \le \sum_j \tau_j - \sum_i u_i \le \tau_n \le T.$$

By symmetry, place the exponent to be increased last. All other factors in (14) remain positive and unchanged. Increasing it by d multiplies the integrand by a factor between 1 and e^{dT}, proving the bound. For $n = 1$ the same assertion is immediate. Limits from distinct nodes within $[0,T]$ prove the confluent version; its denominator remains positive. □

4 Extremal tests and uniqueness

4.1 Finite moment cones at the critical time

For $N \geq 2$, let I_N be the set of real parameters s for which some $w_N \in \mathcal{M}_+([0,T])$ satisfies

$$\int e^{\lambda_j t}\, dw_N(t) = m_j(s) := \frac{e^{\lambda_j T} - s}{\lambda_j}, \qquad 1 \leq j \leq N. \tag{15}$$

These auxiliary parameters may be negative; the original ratio remains $r > 0$. The cone of positive moment vectors is closed because its first coordinate bounds mass, after which weak-star compactness applies. It is full dimensional by Lemma 3.1. The vector $m(1)$, represented by dt, is interior: the Gram matrix

$$G_{ij} = \int_0^T e^{(\lambda_i+\lambda_j)t}\, dt$$

is positive definite, so small perturbations of $m(1)$ have positive density $1 + \sum_i d_i e^{\lambda_i t}$. The first two moments give

$$\frac{9e^{pT} - e^{9pT}}{8} \leq s \leq e^{pT}. \tag{16}$$

For the lower bound, integrate $e^{9pt} - e^{pt} \geq 0$ against w_N and use (15); the upper bound follows from the first moment. Consequently $I_N = [s_N^-, s_N^+]$ is a bounded closed interval containing r, with $s_N^- < 1 < s_N^+$. These intervals are nested.

Lemma 4.1. *There are nonzero functions $q_N(t) = \sum_{j=1}^N c_{j,N} e^{\lambda_j t}$, nonnegative on $[0,T]$, and parameters*

$$s_N = s_N^- \quad (r < 1), \qquad s_N = s_N^+ \quad (r > 1),$$

such that $s_N \to r$. They can be normalized so that

$$J_N := \int_{-\infty}^0 q_N(t)\, dt = \begin{cases} -1, & r < 1, \\ 1, & r > 1, \end{cases} \tag{17}$$

$$\int_0^T q_N(t)\, dt = |1 - s_N| \leq C_T, \tag{18}$$

$$\int_{[0,T]} q_N(t)\, dw(t) = \delta_N \longrightarrow 0 \quad \textit{for every critical scalar measure } w, \tag{19}$$

where $\delta_N = r - s_N$ for $r < 1$, and $\delta_N = s_N - r$ for $r > 1$.

Proof. The endpoint measures of (15) have uniformly bounded mass by (16). Their weak-star subsequential limits represent every odd moment at the limiting endpoint parameters $s^- \leq r \leq s^+$. If $r < 1$ and $s^- < r$, mix the representation at s^- with dt using $\theta = (1-r)/(1-s^-) \in (0,1)$:

$$\theta w^- + (1-\theta)dt.$$

It represents r and has positive interval density, contradicting Lemma 2.2. This argument remains valid when $s^- < 0$. For $r > 1$ and $s^+ > r$, the same contradiction uses $\theta = (r-1)/(s^+ - 1)$. Thus the chosen endpoints tend to r.

The moment vector $m(s_N)$ lies on the boundary of the closed full-dimensional cone. A supporting hyperplane gives a nonzero coefficient vector c_N with $q_N \geq 0$ on $[0,T]$ and $\int q_N\, dw_N = 0$ for an

endpoint measure w_N. For completeness, take points z_ℓ outside the cone tending to this boundary point, and let p_ℓ be their nearest cone points. Minimizing squared distance gives

$$\left\langle \frac{p_\ell - z_\ell}{\|p_\ell - z_\ell\|}, c - p_\ell \right\rangle \geq 0 \quad (c \text{ in the cone}).$$

Here $p_\ell \to m(s_N)$. A convergent subsequence of unit normals gives a nonzero normal nonnegative on the cone and zero at $m(s_N)$, by testing zero and $2m(s_N)$. Testing the exponential evaluation vectors gives the asserted q_N.

Since every $\lambda_j > 0$, its negative-time integral is $J_N = \sum_j c_{j,N}/\lambda_j$. The endpoint moments imply

$$0 = \int q_N \, dw_N = \int_0^T q_N(t)\, dt + (1 - s_N) J_N. \tag{20}$$

The first integral is positive because q_N is nonzero and nonnegative on a nontrivial interval. Hence $J_N \neq 0$ has the stated sign, and positive rescaling gives (17) and (18). Substituting the full moments of any critical w into the same functional gives (19). □

Lemma 4.2. *Each q_N in Lemma 4.1 has exactly $N-1$ real zeros counting multiplicity. All lie in $[0,T]$. Its coefficients are nonzero.*

Proof. The endpoint measure w_N is nonzero, since otherwise its first two moments would force $s_N = e^{pT} = e^{9pT}$. Its support lies in the finite zero set of q_N. Let n be the number of its atoms, and let $i_0, i_T \in \{0,1\}$ indicate endpoint atoms. Nonnegativity of q_N requires at least

$$M = 2n - i_0 - i_T \leq N - 1$$

zeros, using double interior roots and at least simple endpoint roots.

Suppose $r < 1$. If $M + i_0 \leq N - 1$, Lemma 3.1 constructs an interpolant P in the first $M + i_0 + 1$ odd exponentials with double roots at all atoms except a simple root at T, when present. Choose its sign positive on $(-\infty, T)$. Both integrals in

$$\int_0^T P(t)\, dt + (1 - s_N) \int_{-\infty}^0 P(t)\, dt$$

are positive, whereas the finite moments identify this expression with $\int P \, dw_N = 0$. Therefore $M + i_0 > N - 1$, forcing $i_0 = 1$ and $M = N - 1$.

Suppose $r > 1$. If $M + 1 - i_0 \leq N - 1$, construct an interpolant with a simple root at zero, double interior roots and a simple root at T, when present. It uses the first $M + 2 - i_0$ odd exponentials. Choose it positive on $(0, T)$; it is negative on $(-\infty, 0)$. Since $1 - s_N < 0$, the same expression is strictly positive and again contradicts $\int P \, dw_N = 0$. Thus $i_0 = 0$ and $M = N - 1$.

In both cases the global zero bound is exhausted by these roots. Their multiplicities are exact, there are no additional real roots, and the cofactor conclusion of Lemma 3.1 gives all N nonzero coefficients. □

4.2 A coefficient that survives the limit

The nonnegative tests have bounded integrals on the physical interval, while their negative-time integrals remain equal to 1 or -1. We use these two facts to find one fixed coefficient that does not vanish as N increases.

Lemma 4.3. *Along a subsequence of the tests q_N, there are a fixed index m and $c > 0$ such that $|c_{m,N}| \geq c$ eventually.*

Proof. Put $E = e^{pT} > 1$ and $x = e^{pt}$. Write

$$q_N(t) = x f_N(x), \qquad f_N(x) = \sum_{j=1}^{N} c_{j,N} x^{\beta_j}, \qquad \beta_j = (2j-1)^2 - 1 = 0, 8, 24, \ldots.$$

The positive exponents have summable reciprocals and gaps at least 8. On $[1, E]$, $f_N \geq 0$ and

$$\int_1^E f_N(x)\, dx = p \int_0^T q_N(t)\, dt \leq p C_T. \tag{21}$$

We use three results of Borwein–Erdélyi [4]. For $0 = \gamma_0 < \gamma_1 < \cdots$ with $\sum_{i \geq 1} \gamma_i^{-1} < \infty$, their Theorem 5.6 bounds the supremum of a finite polynomial in these powers on $[0, \rho]$ by its L^q norm on any measurable $A \subset [\rho, 1]$ of fixed positive measure. Their Theorem 3.2, when $\gamma_1 \geq 1$, bounds the derivative on $[0, 1 - \eta]$ by the supremum on $[0, 1]$, with a constant independent of the number of terms. Their Theorem 6.1, under a positive lower gap bound, gives a convergent expansion in these powers inside $[0, R)$ for every uniform limit on $[0, R]$.

For each $R < E$, choose $\max(1, R) < R' < E$. Apply the first result with $q = 1$ to $f_N(Ez)$, on $A = [R'/E, 1]$. Equation (21) then bounds $\|f_N\|_{[0,R']}$ uniformly in N. For every fixed k, the derivative $f_N^{(k)}$ belongs to the power space with exponents 0 and $\{\beta_j - k : \beta_j > k\}$. The positive exponents are integers at least one, and their reciprocals are summable: for $\beta_j \geq 2k$, $(\beta_j - k)^{-1} \leq 2/\beta_j$, and only finitely many terms precede this tail. Terms with $\beta_j = k$ are constant, and terms with $\beta_j < k$ vanish. Successive derivative estimates on nested radii between R and E therefore bound every fixed derivative on $[0, R]$.

The bound for the next derivative gives equicontinuity. Arzelà–Ascoli and a diagonal subsequence over derivative orders and an exhaustion of $[0, E)$ yield locally uniform limits for every derivative. Passing through the fundamental theorem of calculus identifies them as the derivatives of one function f. The closure theorem gives $f(x) = \sum_j c_j x^{\beta_j}$ on every $[0, R)$, $R < E$. Since the exponents are integers, this is analytic at zero: convergence at a larger positive y bounds $|c_j| y^{\beta_j}$, giving absolute uniform convergence on smaller complex disks. Taylor coefficients are consequently unique, and

$$c_{j,N} = \frac{f_N^{(\beta_j)}(0)}{\beta_j!} \longrightarrow c_j \quad \text{for each fixed } j.$$

Because $E > 1$, convergence is uniform on $[0, 1]$. The change of variables in the negative-time integral gives

$$\frac{1}{p} \int_0^1 f_N(x)\, dx = J_N = \pm 1.$$

Thus $f \neq 0$, and some $c_m \neq 0$. Along this single subsequence $|c_{m,N}| \geq c > 0$ eventually. The subsequence and m depend only on the finite tests at the fixed r, T. □

4.3 Inserted exponents and the two rigidity statements

Fix the subsequence and index m from Lemma 4.3. Recall that every q_N has $N - 1$ roots and that its cost $\int q_N\, dw = \delta_N$ tends to zero for every critical w. For any fixed

$$\lambda_m < \mu < \nu_m \tag{22}$$

and $N > m$, replace λ_m in q_N by μ, retain all its roots with their multiplicities, and preserve the coefficient of λ_N. Denote the new interpolant by P_N and its μ coefficient by d_N. The moved exponent stays below λ_{m+1}, and moves toward the preserved highest exponent. Lemma 3.2 gives

$$|P_N(t)| \le q_N(t) \qquad (0 \le t \le T). \tag{23}$$

Lemma 4.4. *For every fixed μ in (22), $|d_N| \ge c_\mu > 0$ eventually along the chosen subsequence.*

Proof. Let D_j be the positive minor of the $(N-1)$-by-N confluent root matrix with column j omitted. An interpolant with last coefficient c_N has coefficients $c_j = c_N(-1)^{N-j}D_j/D_N$. Replacing column m leaves D_m unchanged. Since the last coefficient is preserved,

$$\frac{d_N}{c_{m,N}} = \frac{D_N^{\text{old}}}{D_N^{\text{new}}} > 0. \tag{24}$$

These are square minors omitting the highest exponent. Their node factors in the confluent normalization agree and cancel. Their exponent Vandermondes satisfy

$$\frac{V_{\text{new}}}{V_{\text{old}}} = \prod_{j<m} \frac{\mu-\lambda_j}{\lambda_m-\lambda_j} \prod_{m<j\le N-1} \frac{\lambda_j-\mu}{\lambda_j-\lambda_m} \le A_m(\mu) := \prod_{j<m} \frac{\mu-\lambda_j}{\lambda_m-\lambda_j}.$$

All factors are positive, and every factor in the second product is less than one. The normalized determinant ratio is at most $e^{(\mu-\lambda_m)T}$ by Lemma 3.3. Therefore

$$|d_N| \ge \frac{e^{-(\mu-\lambda_m)T}}{A_m(\mu)} |c_{m,N}| \ge c_\mu > 0. \tag{25}$$

The empty product for $m = 1$ is one. The constants may depend on the fixed μ, m, T, but not on N. □

We will use the following elementary uniqueness fact. If a finite signed measure η on $[0,T]$ satisfies $\int e^{zt}\,d\eta(t) = 0$ for all real z in a nonempty open interval, then $\eta = 0$. At an interior point z_0, termwise differentiation is justified by the exponential series and the finite-variation majorant $e^{|z_0|T}e^{|z-z_0|T}\,\|\eta\|_{\text{TV}}$. All derivatives vanish, so $\int t^k e^{z_0 t}\,d\eta = 0$ for every $k \ge 0$. Uniform approximation by real polynomials gives zero integral against every continuous function for the weighted measure $e^{z_0 t}\eta$. Uniqueness of a finite Radon measure from continuous tests, followed by multiplication by $e^{-z_0 t}$, gives the assertion.

Proposition 4.5. *There is exactly one critical scalar measure. Moreover, if w is critical, $|h| \le w$, and $\int e^{\nu_j t}\,dh(t) = 0$ for every $j \ge 1$, then $h = 0$.*

Proof. For two critical measures let $v = (w_1 + w_2)/2$ and $\sigma = (w_1 - w_2)/2$. Then v is critical, $|\sigma| \le v$, and all odd exponential moments of σ vanish. Every term of P_N except its μ term belongs to that odd family. Equations (19), (23) and (25) give

$$|d_N| \left| \int e^{\mu t}\,d\sigma(t) \right| = \left| \int P_N(t)\,d\sigma(t) \right| \le \int q_N(t)\,dv(t) = \delta_N \longrightarrow 0.$$

Thus the exponential transform of σ vanishes at every fixed μ in (λ_m, ν_m), and the uniqueness fact gives $\sigma = 0$.

For the even family, keep the coefficient d_N at μ and move the remaining exponents toward it:

$$\lambda_j \mapsto \nu_j \quad (j < m), \qquad \lambda_j \mapsto \nu_{j-1} \quad (j > m).$$

Perform the lower moves in increasing index order and the upper moves in decreasing index order. The interlacing $\lambda_j < \nu_j < \lambda_{j+1}$ ensures that no exponent is crossed. At the central gap, $\nu_{m-1} < \mu < \nu_m$ (with the lower inequality omitted for $m = 1$). Repeated use of Lemma 3.2 produces an interpolant Q_N with exponents $\mu, \nu_1, \dots, \nu_{N-1}$, the same coefficient d_N at μ, and $|Q_N| \le |P_N| \le q_N$ on $[0, T]$. For the signed measure in the statement,

$$|d_N| \left| \int e^{\mu t}\, dh(t) \right| = \left| \int Q_N(t)\, dh(t) \right| \le \int q_N(t)\, dw(t) = \delta_N \longrightarrow 0.$$

Its transform also vanishes on (λ_m, ν_m), hence $h = 0$. The subsequence was fixed before choosing either measure or μ; each fixed μ uses its own positive lower bound independent of N. □

Proof of Theorem 1.1. Lemma 2.1 gives an admissible pair at the original bounded minimum time. The parity map (5) sends every such pair to a critical scalar measure w and a signed h dominated by w, with every even moment zero. Proposition 4.5 makes w unique and forces $h = 0$. Its inverse is the unique pair (bw, bw). All measure integrals and test inequalities were on the closed interval, so the conclusion includes initial and terminal atoms. □

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